React.JS Training

2021

Roadmap

- Module 0: JavaScript fundamentals for front-end development
- Module 1: React.JS Fundamentals and Concepts
- Module 2: Redux Fundamentals and Concepts
- Module 3: Development of Demo Project

Options for self-learning

Learn React.JS

- Week 1 - Day 0
- Week 1 - Day 1
- Week 1 - Day 2
- Week 1 - Day 3
- Week 1 - Day 4
- Week 1 - Day 5

Some concepts to review on self-basis

Week 2, 3 - Demo Project (Calories Management System)

- Business Requirements
- Functional Requirements
  - Milestone 1 (5 Days)
  - Milestone 2 (5 Days)

Evaluations

Evaluation Criteria

Roadmap

This training program is being devised with the view of providing an extremely hands-on approach to the developer learning React.JS. It will be divided into the following modules:

Module 0: JavaScript fundamentals for front-end development

In this module, revise your JavaScript concepts - Discussion and Development - 1 day

Module 1: React.JS Fundamentals and Concepts

At the end of this module, the learner must have conceptual and practical knowledge of React.JS fundamentals - Discussion and Development - 3 days

Module 2: Redux Fundamentals and Concepts

At the end of this module, the learner must have conceptual and practical knowledge of Redux fundamentals - Discussion and Development - 2 days

Module 3: Development of Demo Project

Development of a simple project that includes state management, authorization, authentication, routing, pagination - Initial Discussions, Navigation and Development - 2 weeks

Options for self-learning

If you're planning to self-learn react then you can choose either option

1. Register yourself in below codecademy courses and complete all the exercises by yourself. After the completion of both courses your module 1 and 2 will be completed and mark the content mentioned in week 1 of this document completed which you already covered in codecademy courses and go through the remaining content from the reference links provided with each topic.
    a. https://www.codecademy.com/learn/react-101
    b. https://www.codecademy.com/learn/learn-redux

2. First complete week 1 as per outline mentioned in week 1 of this document and then complete above codecademy courses for better understanding of react concepts.

Learn React.JS

We'll have **daily discussions** initially & **practical progress** will be **tracked via GitHub repositories**. We will focus on a hands-on approach and learning more through actual development and concluding the learning phases with coding challenges and discussions.

Apart from this training program, if you are interested to study the internals of JavaScript on your own, then "**You Don't Know JavaScript (YDKJS)**" Book Series (https://github.com/getify/You-Dont-Know-JS) is a great learning resource on this subject matter.

Rule of thumb: Consider Google and Stack Overflow as your best friends.

Note: We will use **ES6**, **Webpack 4** and **React v17.0.2** for development purposes. **WebStorm** is the best JavaScript IDE for development, but it's paid. If you can arrange its license, then it's highly recommended. Other option is Visual Studio Code (VSCode), it is free and can be used for code editing purposes as it provides a better Developer Experience (DX) for JS Developers.

Week 1 - Day 0

1. Revise your basic JavaScript concepts before getting started with React learning. You can choose any reference site for this purpose or just go through the content of Introduction to JavaScript course of codecademy.
    a. https://developer.mozilla.org/en-US/docs/Web/JavaScript
    b. Introduction to JavaScript

Week 1 - Day 1

1. Learning Main Concepts of React.JS (Components, Props, State, Events)
    a. https://reactjs.org/docs/hello-world.html
    b. https://reactjs.org/docs/introducing-jsx.html
    c. https://reactjs.org/docs/rendering-elements.html
    d. https://reactjs.org/docs/components-and-props.html
    e. https://reactjs.org/docs/state-and-lifecycle.html
    f. https://reactjs.org/docs/handling-events.html

Week 1 - Day 2

1. Learning Main Concepts of React.JS (Conditional Rendering, Lists, Forms, State Lifecycle)
    a. https://reactjs.org/docs/conditional-rendering.html

    b. https://reactjs.org/docs/lists-and-keys.html
    c. https://reactjs.org/docs/forms.html
    d. https://reactjs.org/docs/lifting-state-up.html
    e. https://reactjs.org/docs/composition-vs-inheritance.html
    f. https://reactjs.org/docs/thinking-in-react.html
2. Learning Virtual DOM and Reconciliation Concepts
    a. https://www.codecademy.com/articles/react-virtual-dom
    b. https://reactkungfu.com/2015/10/the-difference-between-virtual-dom-and-dom/
    c. https://reactjs.org/docs/reconciliation.html

Week 1 - Day 3

1. Setup local environment for basic programming practice in React.JS
    a. https://docs.npmjs.com/downloading-and-installing-node-js-and-npm
    b. https://appdividend.com/2017/03/28/beginners-guide-to-setup-es6-developmentenvironment/
2. **Coding Challenge:** Implement React.JS **Tic Tac Toe** that covers all basic concepts by following the official tutorial:
    a. https://reactjs.org/tutorial/tutorial.html

Week 1 - Day 4

1. Learning fundamental concepts of Redux
    a. https://redux.js.org/introduction/three-principles
    b. https://www.smashingmagazine.com/2016/06/an-introduction-to-redux/
    c. http://devguides.io/redux/
2. Learning Basics of React Redux, Redux Thunk
    a. https://react-redux.js.org/introduction/quick-start
    b. https://daveceddia.com/what-is-a-thunk/
    c. https://redux.js.org/advanced/async-actions
3. Setup local environment for basic programming practice with Redux
    a. https://docs.npmjs.com/downloading-and-installing-node-js-and-npm
    b. https://appdividend.com/2017/03/28/beginners-guide-to-setup-es6-developmentenvironment/
4. **Coding Challenge:** Implement a basic webpage in React.JS using Redux that submits a request to a weather API, passes a city as input and displays the list of temperatures expected in the next five days in that city

Week 1 - Day 5

1. Learning fundamental concepts of routing in React
    a. https://reacttraining.com/react-router/web/guides/quick-start
    b. https://github.com/ReactTraining/react-router
    c. https://medium.freecodecamp.org/hitchhikers-guide-to-react-router-v4-a957c6a5aa18
2. Learning fundamental concepts of forms in Redux
    a. https://redux-form.com/8.1.0/
    b. https://github.com/erikras/redux-form
3. Learning about integrating bootstrap in React.JS
    a. https://react-bootstrap.github.io/
4. **Coding Challenge:** Implement a small project using React.JS and Redux containing two separate screens. On the first screen, you will add redux form with validations and on the second page, listings will be displayed for all submitted forms. Firebase can be used to avoid backend development for this task.
    a. https://www.youtube.com/watch?v=mwNATxfUsgI
    b. https://www.youtube.com/watch?v=p4XTMvagQ2Q

Some concepts to review on self-basis

1. Learning Basic ES6 concepts and ES6 Classes
    a. https://hackernoon.com/es6-for-beginners-f98120b57414
    b. https://coursework.vschool.io/es6-basics/
    c. https://scotch.io/tutorials/demystifying-es6-classes-and-prototypal-inheritance
2. Array Helpers and Imports (Named & Default) in ES6
    a. https://hackernoon.com/es6-for-beginners-77bf34bec2d1
    b. https://www.karolisram.com/array-helpers/
    c. http://2ality.com/2014/09/es6-modules-final.html
3. Performing Async operations in ES6 (Promises, Async/Await)
    a. https://hackernoon.com/es6-for-beginners-part-2-ee8a77f7f4c7
    b. https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Promise
    c. http://jamesknelson.com/grokking-es6-promises-the-four-functions-you-need-to-avoid-callback-hell/
    d. http://callbackhell.com/
4. Learning relation of Babel with ES6 and Basics
    a. https://babeljs.io/docs/en/
5. Learning Basics of Webpack
    a. https://webpack.js.org/concepts/
6. Develop a basic ES6 Class (along with GitHub Repository URL) that can request any free Weather API and provide information of either single or multiple cities or return a list of top 10 coldest cities today. Also, try to add features that involve usage of these concepts: **Default Exports**, **Named Exports**, **Destructuring Object and Arrays**, **Inheritance**.
7. Server-Side Rendering
8. Higher Order Components
9. Fragments
10. React Hooks
11. Unit Testing
12. Redux Saga

Week 2, 3 - Demo Project (Calories Management System)

1. A demo application will be assigned to you from React.JS Team @ Square63 2. Requirements will be provided by React.JS Team
3. React.JS Team will act as client for you.
4. You have to clarify all the requirements by asking question to React.JS Team @Square63
5. You have to follow the deadline for giving requirements.
6. You are supposed to implement all concepts in this application according to best of you knowledge
7. Your code will be reviewed frequently by React.JS Team
8. When you are done with this demo application then you will have a discussion with React.JS Team about this application.
9. Usage of boilerplates will be discouraged strictly.

Business Requirements

1. We need to build an application for daily calorie intake of a person.
2. Regular users will be able to view/add/edit/delete meals and set their daily calorie intake limit.
3. If the meal intake is higher than the daily limit, the listings for exceeding meals will turn red. Listings of meals within limit will remain green.
4. Managers can manage their meals only like regular users and also do CRUD operations on users.
5. Admins will be able to delete/edit meals for users, view meals added for all users and manage users through CRUD operations.
6. Users will be able to filter meal listings by name, user, starting date and ending date.
7. You can maintain mock REST API from JSON arrays using following tutorials https://medium.com/backticks-tildes/building-a-mock-rest-api-with-json-server-812ed3e6036a https://my-json-server.typicode.com/

Functional Requirements

Milestone 1 (5 Days)

1. Setup Application with Redux and React Router
2. Sign Up Screen
3. Sign In Screen
4. Create new User Screen
5. Edit User Screen
6. User Details Screen
7. User Listings Screen
8. User Settings Screen (To set daily calorie intake limit)
9. Styling with Bootstrap

Milestone 2 (5 Days)

10. Create new Meal Screen
11. Edit Meal Screen
12. Meal Details Screen
13. Meal Listings Screen
14. Filtering of Meals by Name, Start Date and End Date
15. Labeling of Meals Listing by green/red color as per requirement.
16. Authorization of Routes as per user roles.
17. State Persistence
18. UI Enhancements
19. Deployment

***For user requirements & non-functional requirements discuss with your mentors.**

Evaluations

Evaluations after the end of each module/milestone will be conducted on your progress on the basis of coding challenges and demo project. You will receive feedback based upon the following criteria:

Evaluation Criteria

Productivity

- Given tasks completed by accepting challenges
- Code quality improvement via commit reviews

Responsibility

- Taking ownership of the work
- Reliability & Dependability under pressure

Time management

- Whether given tasks are divided into their ideal time span

Innovation

- New ideas
- Creative approach towards a problem

Learning

- Learning new things everyday
- Sharing your knowledge with others

Work Ethics

- Proper office discipline is observed while working under workload